



How can I help you?



AI-BASED CUSTOMER SUPPORT AUTOMATION SYSTEM

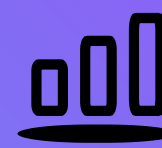
An Intelligent Helpdesk Solution Using Artificial Intelligence and Natural Language Processing

Team Members: Akrishta Ale | Ayush Bhandari | Prajita Bhandari

Unit Coordinator: Farzad Sanati



Faster Responses



Efficient Support



Happier Customers



INTRODUCTION

Customer support teams often face high volumes of repetitive enquiries, leading to delayed responses and inconsistent service quality.

This project develops an AI-Based Customer Support Automation System using Artificial Intelligence (AI) and Natural Language Processing (NLP) to provide fast, accurate and personalised customer support.

The system combines an AI chatbot, automated ticket management, sentiment analysis, classification and intelligent escalation to improve support efficiency while assisting human agents with complex customer issues.



OBJECTIVES



Develop an AI chatbot
to handle common customer enquiries.



Automate ticket management
including ticket creation, classification and prioritisation.



Reduce customer response time
and improve support efficiency.



Implement sentiment analysis
and confidence scoring for intelligent escalation.



Provide role-based access
for customers, support agents and administrators.



Deliver an analytics dashboard
for monitoring ticket trends and system performance.



KEY FEATURES



AI Chatbot
24/7 intelligent support using NLP



Ticket Management
Create | Track | Manage support tickets



AI Analysis
Sentiment analysis | Classification | Priority detection



Smart Escalation
Low confidence or complex issues → Human agents



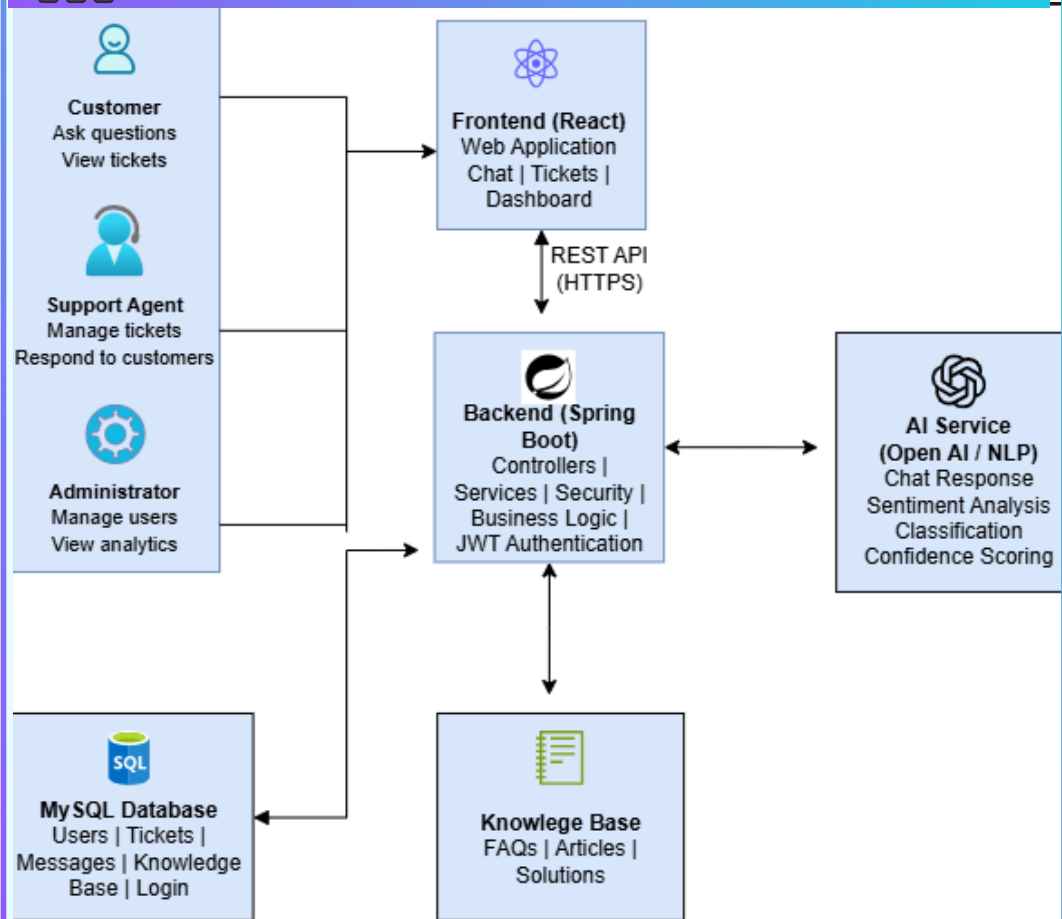
Role-Based Access
Customer | Support Agent | Administrator



Analytics Dashboard
View ticket trends | System performance | Insights



SYSTEM ARCHITECTURE



TECHNOLOGY STACK



CONCLUSION

The AI-Based Customer Support Automation System successfully demonstrate how AI and NLP can be used to automate customer interactions, reduce response time, improve support agent productivity and enhance customer satisfaction. The system provides pratical value for both customers and support teams, with real-time analytics and intelligent escalation for complex issues

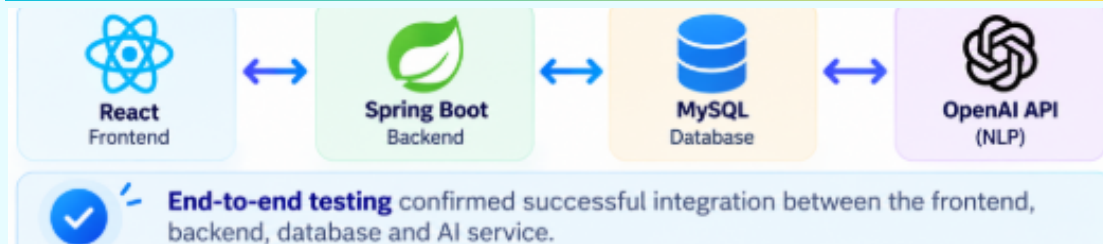


RESULTS & ANALYSIS

AI Function Evaluation



System Integration Results



Functional Testing Results

Feature	Result
User registration and login	✓ Pass
Ticket Creation	✓ Pass
AI chat Respond	✓ Pass
Ticket Classification	✓ Pass
Sentiment Analysis	✓ Pass
Role-Based Access control	✓ Pass
Frontend Backend AI Integration	✓ Pass



FUTURE WORK



ACKNOWLEDGEMENTS

We sincerely thank our UC, Farzad Sanati, for his valuable guidance and support throughout this project. We also acknowledge our team members for their collaboration, dedication, and contributions to the successful development of the project.

INTELLIGENT SUPPORT FOR BETTER TOMORROW

<https://github.com/prajitabhandari1234/AI-Based-Customer-Support-Automation-System>